

SRI BALAJI YELLELA

Fullstack Software Engineer

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SUMMARY

Fullstack Software Engineer with 4 years building distributed backend services in Go and Java across payments and supply chain systems. Works day to day in PostgreSQL, Kubernetes and event-driven messaging, with responsibility for services from design through production support. Comfortable owning latency and reliability targets on systems that other teams depend on. Suited to backend work where correctness under load and clear operational ownership matter.

TECHNICAL SKILLS

- **Languages** Go · Java · Python · C# · TypeScript · JavaScript · **SQL**
- **Backend & Distributed Distributed systems** · **Microservices** · REST APIs · gRPC · Spring Boot · Node.js · .NET · Event streaming
- **Data Stores** PostgreSQL · DynamoDB · SQL Server · Azure SQL Database · CosmosDB · MySQL
- **Infrastructure** Kubernetes · Docker · Terraform · CI/CD pipelines · AWS · GCP · Microsoft Azure
- **Testing** JUnit · Mockito · Jest · Cypress · Selenium · Integration testing
- **Practices** Agile/Scrum · Test-driven development · Object-oriented design · Code review
- **Transferable to target stack** GCP Pub/Sub and AWS SQS (durable message brokers, equivalent role to **Kafka**) · EKS (managed **Kubernetes**)

EXPERIENCE

Stripe | Payments Infrastructure, Bangalore

Aug 2022 - Dec 2023

Software Development Engineer

- Engineered payment authorisation services in **Python** and **Go** that sustained **1M+ daily transactions** at **99.9% uptime**, keeping settlement accurate during peak merchant load.
- Partitioned a monolithic reconciliation job into **microservices** coordinated through **GCP Pub/Sub**, clearing message backlogs that had been breaking settlement for **12 downstream services**.
- Tuned **REST API** request paths and connection pooling to hold **sub-100ms p99 latency** across **200K+ merchant integrations**, which kept checkout responsive as volume grew.
- Chose at-least-once delivery with idempotency keys over exactly-once semantics after weighing operational cost, accepting duplicate handling in exchange for a simpler failure mode the on-call team could reason about.
- Instrumented tracing and alerting across critical payment paths, cutting **mean time to recovery by 35%** and giving merchant support a shared signal to triage against.

Amazon | Supply Chain & Internal Developer Tools, Hyderabad

Jan 2021 - Jul 2022

Software Development Engineer

- Operated cloud-native services on **AWS (Lambda, SQS, S3, EKS)** serving **500K+ daily queries** at **98% availability** for warehouse planners.
- Diagnosed and resolved **PostgreSQL** and **DynamoDB** query bottlenecks that were stalling **150K+ daily inventory operations**, restoring same-day stock accuracy.
- Shipped weekly zero-downtime rolling updates on **Kubernetes**, replacing a maintenance-window release that had blocked fulfilment during deploys.
- Refactored core ordering modules onto consistent **object-oriented design** patterns, reducing module-level defect rates by approximately **30%** and shortening review cycles.
- Presented service design proposals at technical review and guided incoming engineers through the supply chain domain, which shortened their first-commit time.

ACADEMIC PROJECTS

Distributed Key-Value Store with Consensus Protocol | Graduate coursework

Spring 2024

Java, Raft consensus, gRPC, Docker, JUnit

- Implemented a **distributed** key-value store on the **Raft consensus** algorithm with leader election and log replication, holding linearisable reads across a 5-node cluster under simulated network partitions.
- Measured **12K ops/sec** read throughput under load with concurrent failure injection, validating the replication design against the course fault model.

Cloud-Native ETL Pipeline on GCP | Graduate coursework

Fall 2024

Python, Apache Beam, GCP Dataflow, BigQuery, Pub/Sub, Terraform

- Delivered a streaming ETL pipeline moving **100M+ records** from **Pub/Sub** into **BigQuery** with schema validation and failure alerting, holding data loss under **0.1%**.
- Codified the pipeline environment in **Terraform**, cutting full redeployment from 45 minutes to under 4 minutes for each grading iteration.

PROJECTS

Full-Stack Scalable Web Platform

2024 - 2025

React, TypeScript, Spring Boot, PostgreSQL, Docker, JWT, OpenAPI

- Built a **REST API** backend in **Spring Boot** over **PostgreSQL** with a React and **TypeScript** client, holding sub-200ms average response for 200 concurrent users.
- Covered the service layer with **JUnit**, Mockito and RestAssured suites, catching contract regressions before they reached the deployed environment.

EDUCATION

Northeastern University | Boston, MA

Jan 2024 - Dec 2025

Master of Science, Computer Science